A beginner's starting map for Indian space-data platforms

Stuck trying to find satellite data? Start here.

Most students hit the same wall with MOSDAC, NASA Earthdata, or Copernicus — not because they don't understand space, but because no one explains the difference between a portal, an API, and a tool. This page does.

> Built directly from the public documentation of MOSDAC (ISRO), NASA Earthdata, and the Copernicus Data Space Ecosystem — every claim on this page is traceable to a named source.

01

The Data Portal Map

Each platform below contains all three kinds of access — a portal to browse, an API to pull data programmatically, and tools to process it. Knowing which one you're looking at changes what skills you need.

Portal — browse & download API — programmatic access Tool — process once downloaded

Platform	Portal	API	Tool
MOSDAC ISRO, India	Browse Web portal for ocean, atmosphere & land products	Limited Mostly download-based; programmatic access is minimal	Varies Products need external GIS software to open

NASA Earthdata NASA, USA	Browse Search & discovery portal, task-based tutorials	Full Earthdata Search API, well documented	Guided "Data recipes" walk you from raw file to result
Copernicus ESA / EU	Browse Integrated visual data browser	Full Strong API, but assumes you know authentication	Varies Processing tools exist but need setup

02

Task-Based Pathways

Pick the goal closest to yours. Each pathway gives you one concrete next step — not a list of every option.

Find rainfall data | Find land-use data

Goal: **Find satellite rainfall data for your region**

1 **Go to the MOSDAC portal** at mosdac.gov.in

This is the Portal layer — you're browsing, not calling an API yet.

2 **Navigate to "Atmospheric Products"**

This is where rainfall and precipitation datasets are grouped.

3 **Select a rainfall/precipitation dataset and your date range**

Check the Glossary below for what "date range" options actually mean here.

4 **Download in GeoTIFF or NetCDF format**

Both are file formats for geographic data — see Glossary for which one fits your analysis tool.

03

Glossary

The same term means the same thing everywhere on this page. Search or scan below.

Search a term, e.g. 'API' or 'GeoTIFF'...

Portal	A website where you browse and download data by clicking through menus — no coding needed.
API	Application Programming Interface. A way for a program (not a human clicking buttons) to request data directly from a platform.
Tool	Software used after downloading data, to open, view, or process it — separate from the portal or API you got it from.
Endpoint	A specific web address an API responds to — think of it as one "counter" at a service desk that handles one type of request.
Authentication	Proving who you are to an API, usually with a token, before it will respond to your request.
Token	A piece of text that acts like a temporary password, proving to an API that you're allowed to access it.
GeoTIFF	An image file format that also stores exact geographic location — common for satellite imagery.
NetCDF	A file format built for large, multi-dimensional scientific data, such as data that varies by time and location together.
Level-2A product	Satellite data that has already been corrected for atmospheric interference — more ready-to-use than raw (Level-1) data.

04

Sources

Every classification and step on this page traces back to one of these three platforms' own public documentation.

MOSDAC mosdac.gov.in ↗
Indian Space Research Organisation (ISRO)

NASA Earthdata Tools earthdata.nasa.gov ↗
National Aeronautics and Space Administration

Copernicus Data Space Ecosystem dataspace.copernicus.eu ↗
European Space Agency / European Union

This page does not host, process, or provide live access to satellite data. It explains how to reach the three platforms above and does not replace or compete with their official documentation.

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